

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 17

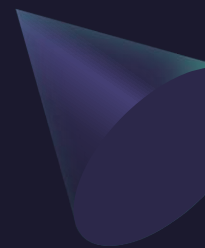


Any Reminders? Any Questions?

- Exam on Friday (discussion) covers 15.1-15.3 + 15.6-15.9
- See email message with resources
 - SP26 Practice Exam/Problems!
- Review during Thursday's class
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!



“Calculus 3”

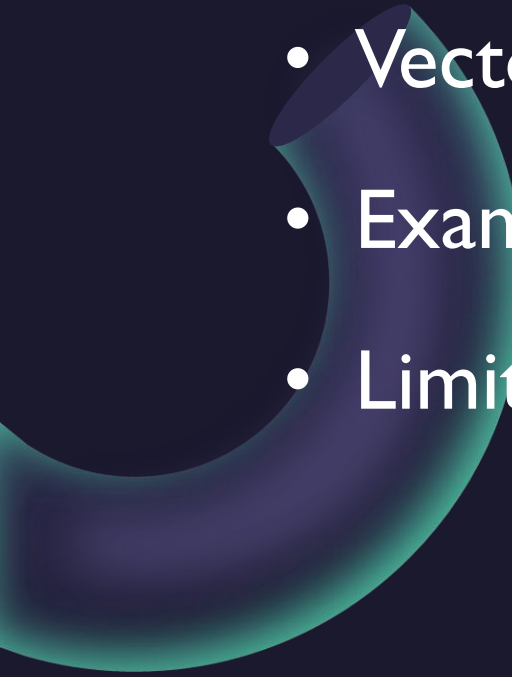
Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Vector Functions and Space Curves



Today – Vector Functions and Space Curves

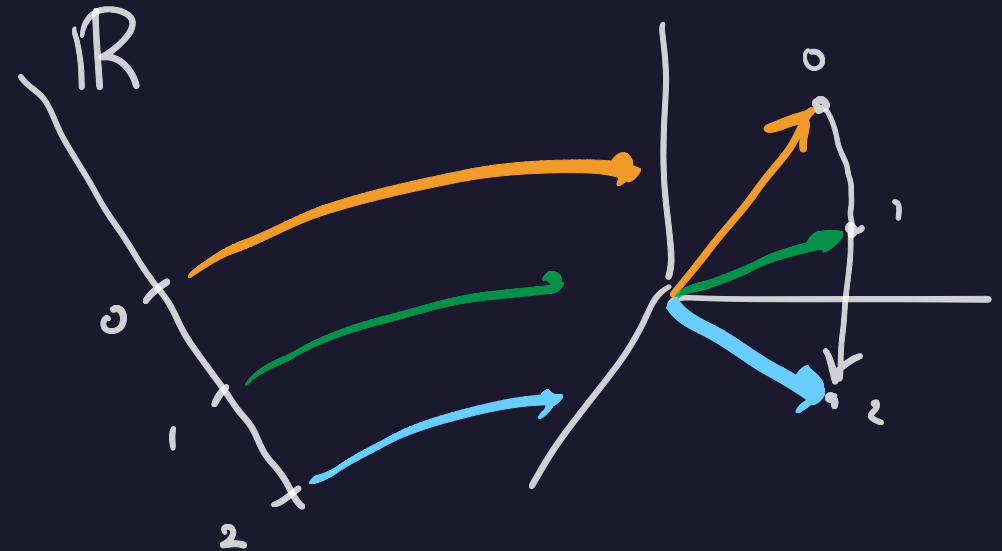
- 
- Vector valued functions
 - Examples
 - Limits and Continuity

What is a “Vector-valued Function”?

$$f: \mathbb{R} \longrightarrow \mathbb{R}^3$$

$$t \longmapsto (f(t), g(t), h(t))$$

$$f: [a, b] \longrightarrow \mathbb{R}^3$$



Example: Describe the image of the function given by

$$r(t) = (1-t, 2, -1+t)$$

$$r: \mathbb{R} \rightarrow \mathbb{R}^3$$

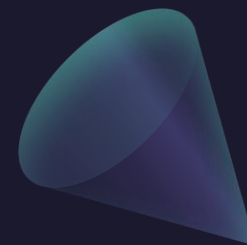
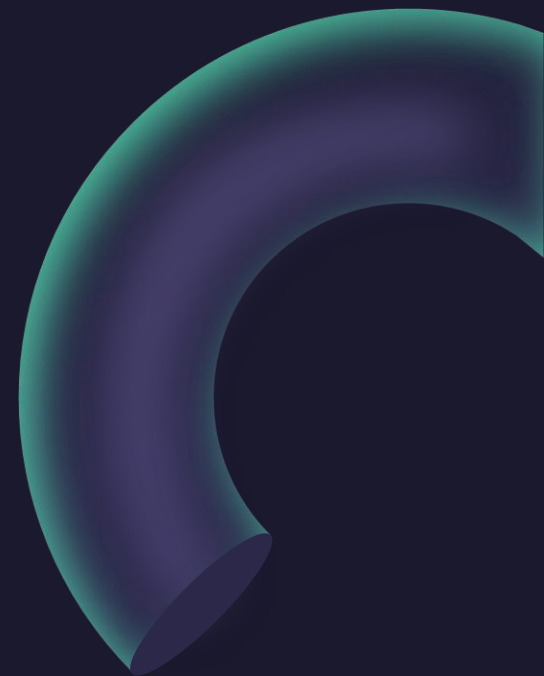
$$t \mapsto (1-t, 2, -1+t)$$

vector form

param
form

$$\begin{cases} x = 1-t \\ y = 2 \\ z = -1+t \end{cases}$$

$$\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$



Example: Describe the image of the function given by

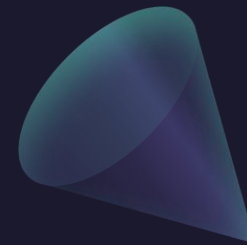
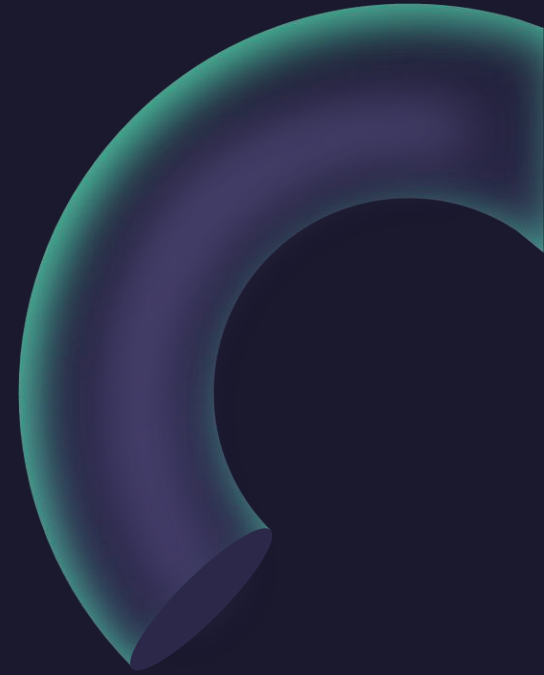
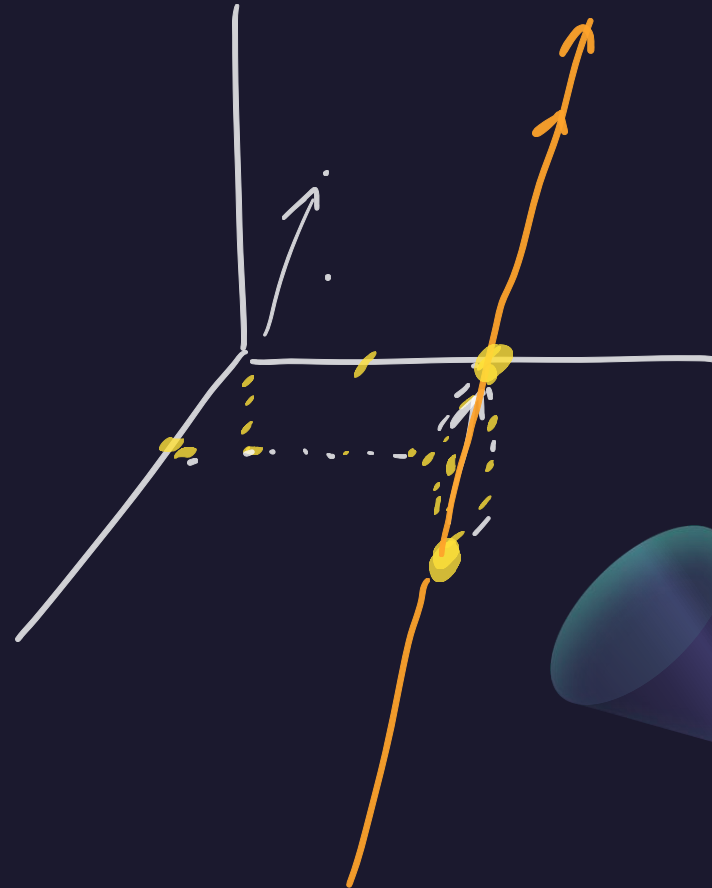
$$r(t) = (1+t, 2+5t, -1+6t)$$

$$(1-t, 2, -1+t)$$

$$\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$r(0) = (1, 2, -1)$$

$$r(1) = (0, 2, 0)$$



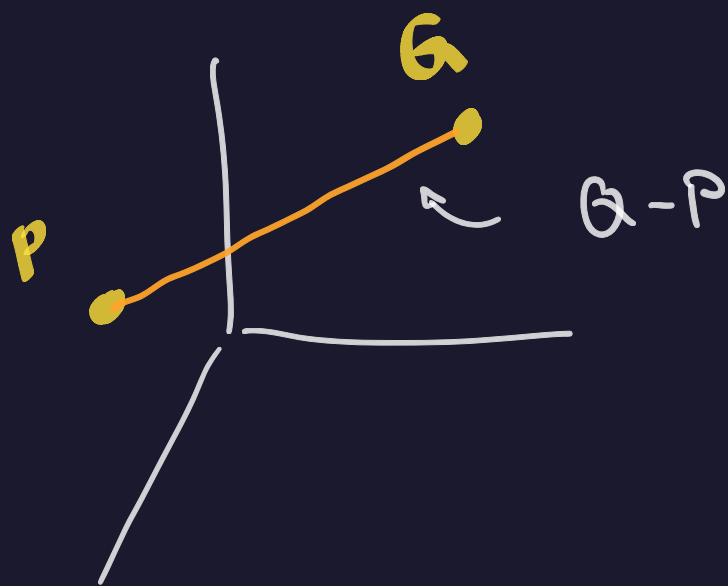
Example: Find a vector equation and parametric equation for the line segment that joins $P = (1, -3, 2)$ and $Q = (2, -1, 3)$.

vector
valued
function

$$r(t) = (1-t)P + tQ$$

$$r: [0, 1] \rightarrow \mathbb{R}^3$$

0 $\mapsto$ P
1 $\mapsto$ Q



$$= P + t(Q-P)$$

$$= \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

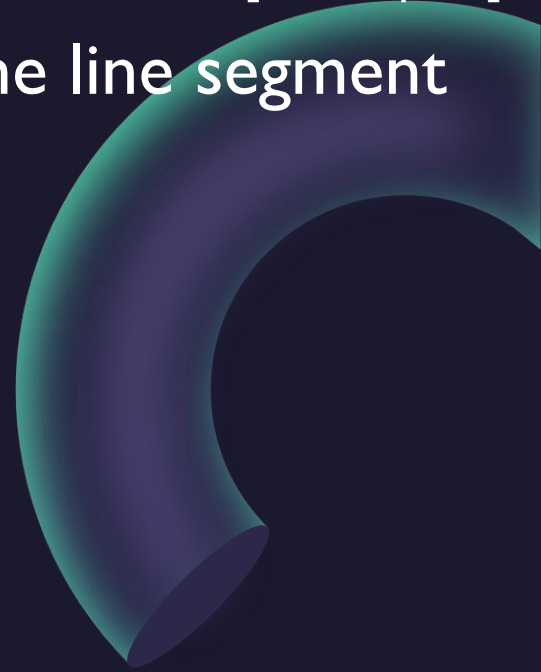
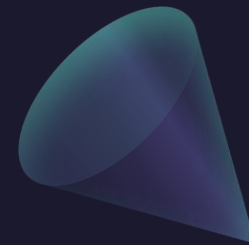
or

$$\begin{cases} x = 1 + t \\ y = -3 + 2t \\ z = 2 + t \end{cases}$$

vector
eq'n form

parametric
form

Example: Find a vector equation and parametric equation for the line segment that joins $P = (1, -3, 2)$ and $Q = (2, -1, 3)$.

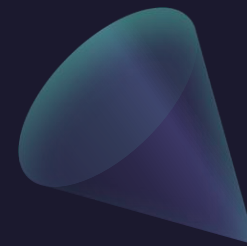
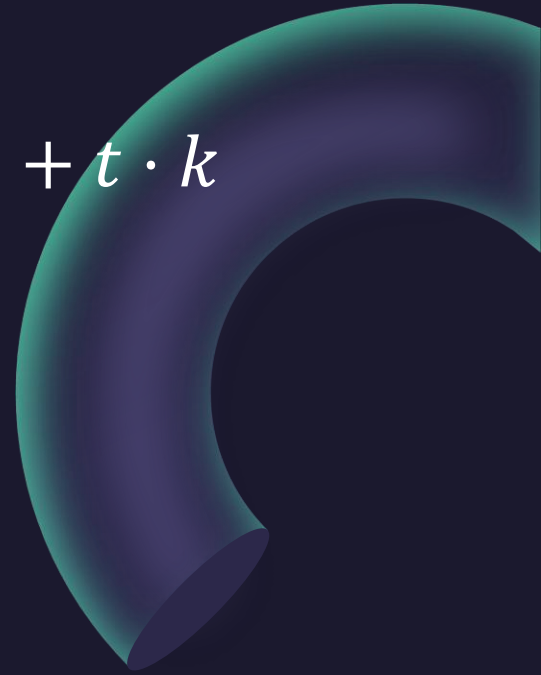
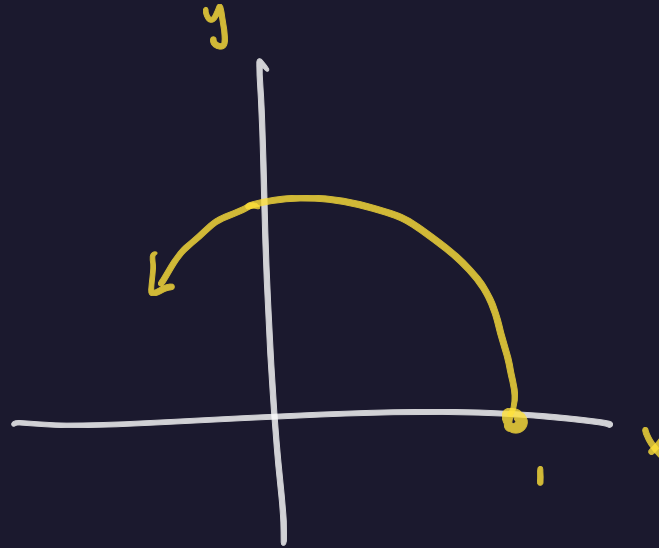


Example: Describe the image of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

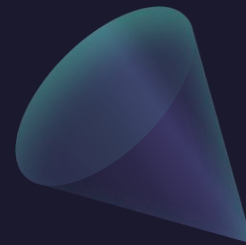
Projection onto xy -plane :

$$\tilde{r}(t) = (\cos t, \sin t)$$

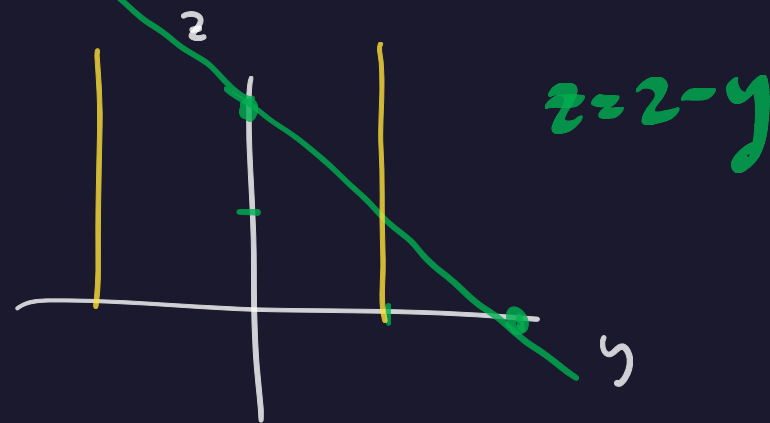


Example: Describe the image of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

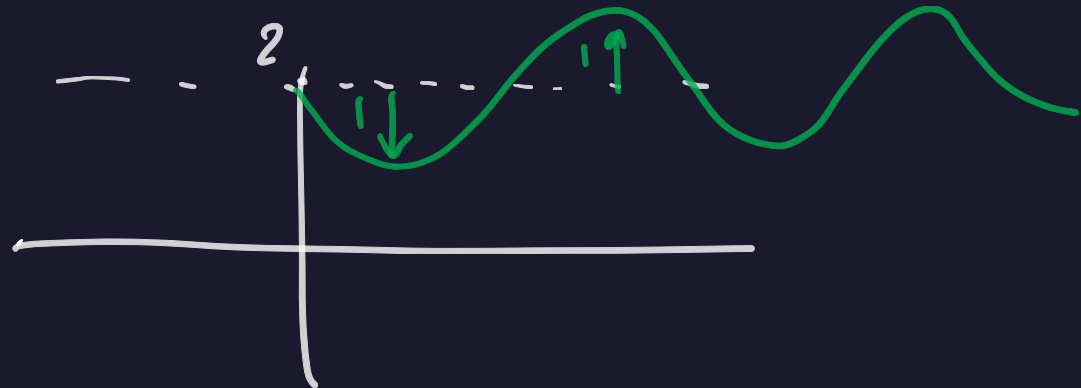


Example: Find a vector equation that represents the curve at the intersection of $x^2 + y^2 = 1$ and the plane $y + z = 2$.

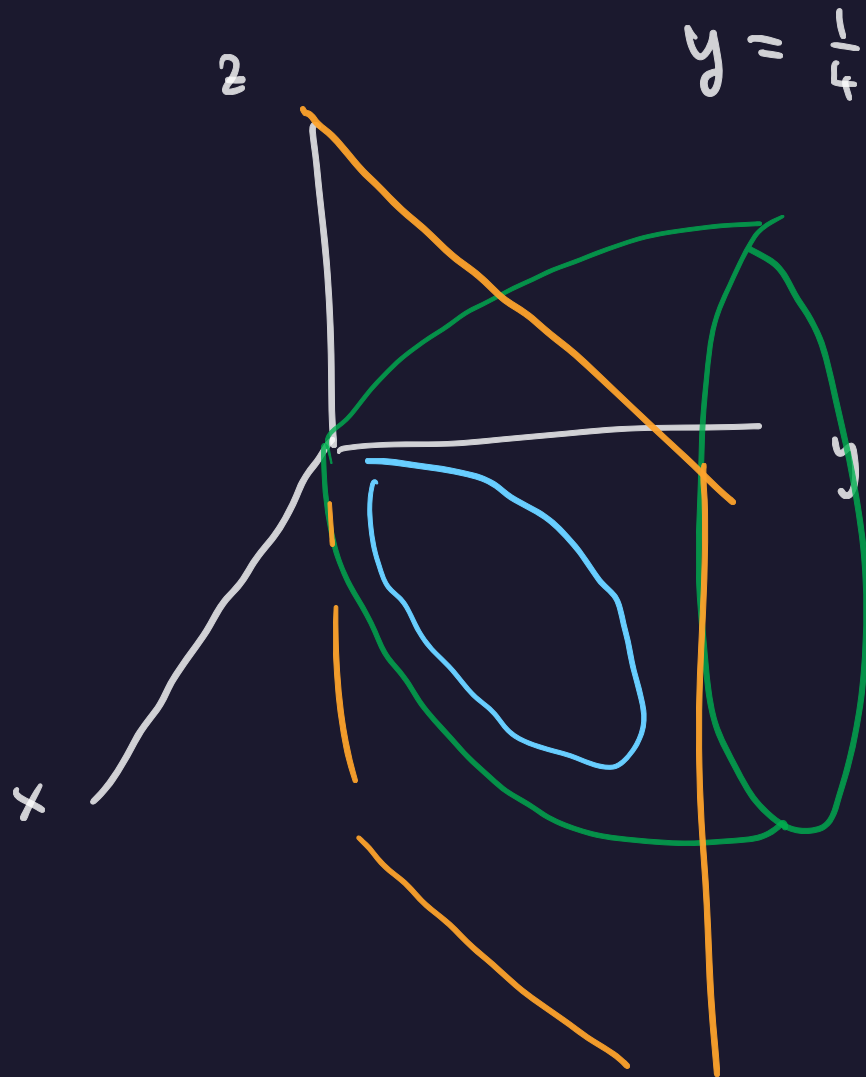


1) parametrize $x^2 + y^2 = 1$
2) $y + z = 2 \rightarrow z = 2 - y$
 $(\cos t, \sin t)$ parametrizes $x^2 + y^2 = 1$
 $\Rightarrow \boxed{(\cos(t), \sin(t), 2 - \sin t)}$





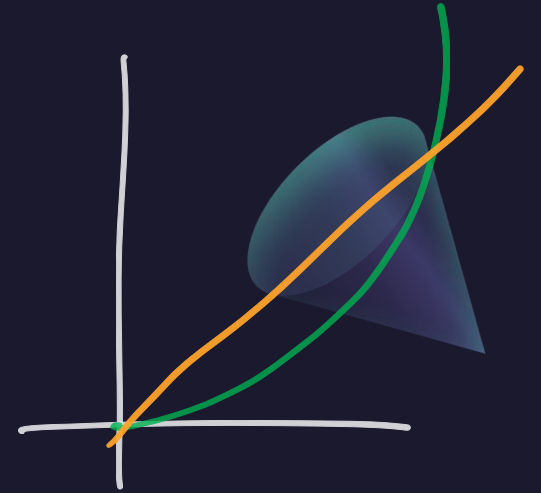
Example: Find a vector equation that represents the curve at the intersection of $4y = x^2 + z^2$ and the plane $y = x$.



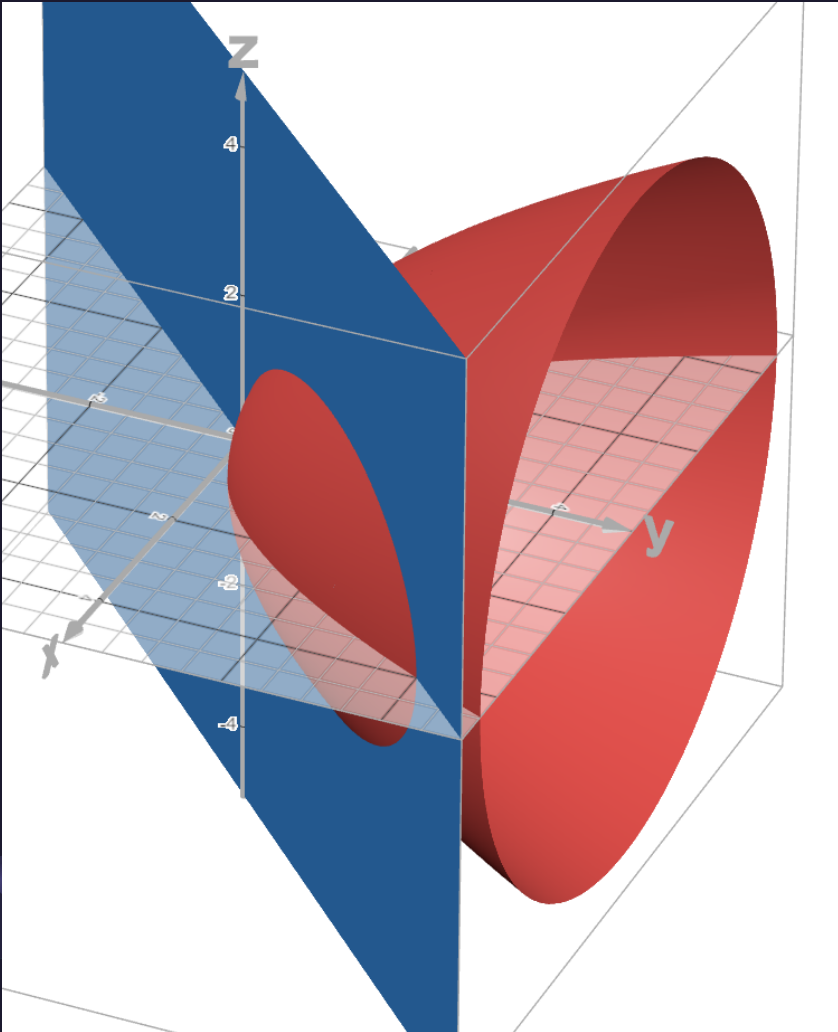
$$z=0$$

$$y = \frac{x^2}{4}$$

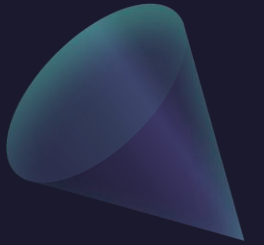
$$y = x$$



Example: Find a vector equation that represents the curve at the intersection of $4y = x^2 + z^2$ and the plane $y = x$.



$$\begin{cases} 4y = x^2 + z^2 \\ y = x \end{cases} \quad \begin{cases} 4x = x^2 + z^2 \\ y = x \end{cases}$$



$$\begin{aligned} 4x &= x^2 + z^2 \rightarrow x^2 - 4x + z^2 = 0 \\ x^2 - 4x + 4 + z^2 &= 4 \\ (x-2)^2 + z^2 &= 4 \end{aligned}$$

$$\begin{aligned} x-2 &= 2\cos t \\ z &= 2\sin t \end{aligned}$$

$$y = 2 + 2\cos t$$

$$\boxed{r(t) = (2 + 2\cos t, 2 + 2\cos t, 2\sin t)}$$

Limits and Continuity of Vector-valued Functions

1 If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle$$

provided the limits of the component functions exist.

A vector function $\mathbf{r}$ is **continuous at a** if

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$$

Example: Find the limit as t goes to 0:

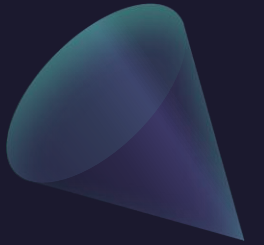
$$\lim_{t \rightarrow 0} \left(1 + t^3, te^{-t}, \frac{\sin(t)}{t} \right)$$

$$= \left(\lim_{t \rightarrow 0} 1 + t^3, \lim_{t \rightarrow 0} te^{-t}, \lim_{t \rightarrow 0} \frac{\sin t}{t} \right)$$

$$= (1, 0, 1)$$

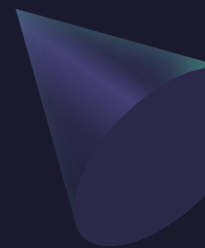
$$\left\{ te^{-t} = \frac{t}{e^t} \rightarrow 0 \right\}$$

Questions?

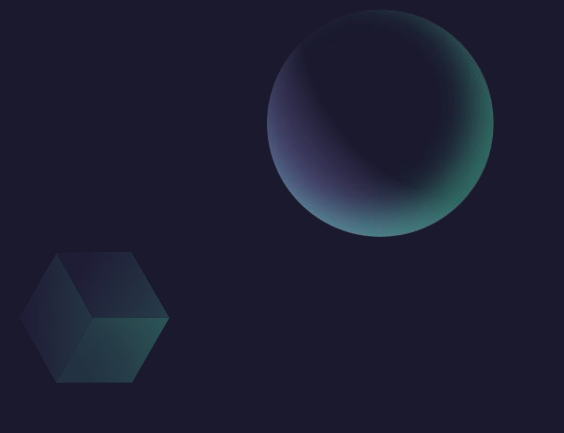




ALVARO: Start the recording!



“Calculus 3”

A sphere, a cube, and a cone are positioned in the upper right area of the slide. The sphere is at the top right, the cube is below it and to the left, and the cone is further down and to the left.

Multi-Variable Calculus

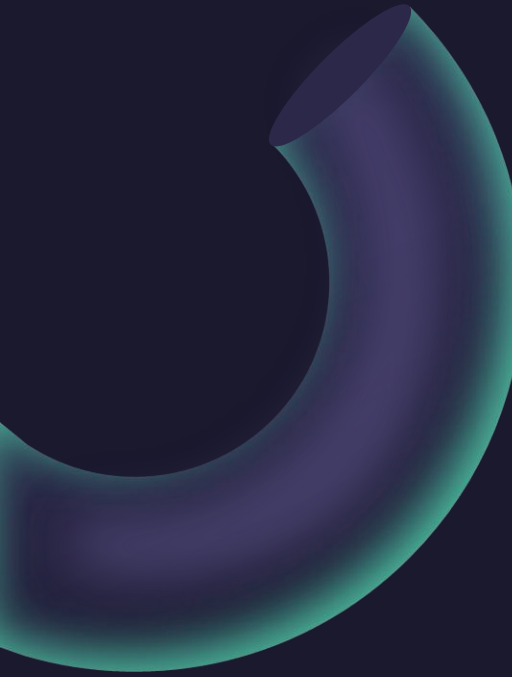
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Derivatives and Integrals Of Vector Functions

A cone and a torus are positioned in the lower right area of the slide. The cone is located between the 'Multi-Variable Calculus' and 'Derivatives and Integrals' text. The torus is a large, thick ring shape on the right side of the slide.

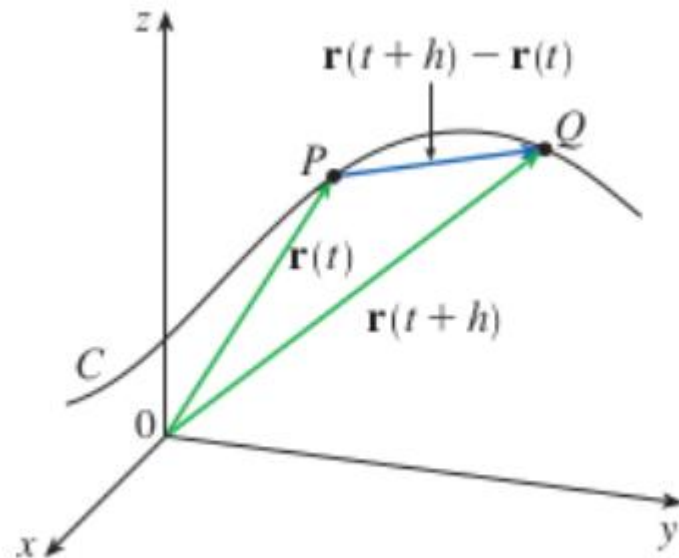
A semi-transparent teal sphere is positioned behind the word 'Today', and a small teal cube is located to its left.

Today – Derivatives and Integrals of Vector Functions

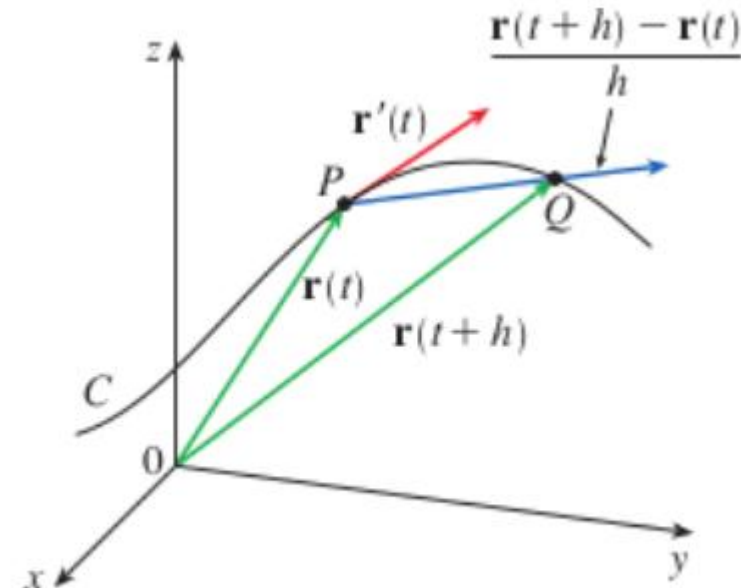
- 
- A teal-colored torus (donut shape) is located on the left side of the slide, partially cut off by the edge.
- Derivatives of vector valued functions
 - Differentiation rules
 - Integrals

What is a derivative of a “Vector-valued Function”?

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$



(a) The secant vector $\overrightarrow{PQ}$



(b) The tangent vector $\mathbf{r}'(t)$

What is a derivative of a “Vector-valued Function”?

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t) \mathbf{i} + g(t) \mathbf{j} + h(t) \mathbf{k}$, where f , g , and h are differentiable functions, then

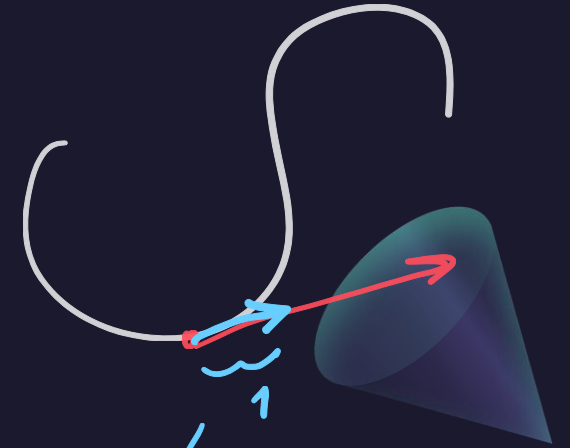
$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle = f'(t) \mathbf{i} + g'(t) \mathbf{j} + h'(t) \mathbf{k}$$

What is a derivative of a “Vector-valued Function”?

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

A unit vector that has the same direction as the tangent vector is called the **unit tangent vector $\mathbf{T}$** and is defined by

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$



Example: Find the derivative of the function given by

$$r(t) = (1 + t, 2 + 5t, -1 + 6t)$$

and find the unit tangent vector at $t = 2$.

$$\frac{d}{dt} r(t) = r'(t) = (1, 5, 6)$$

$$\sqrt{1^2 + 5^2 + 6^2} = \sqrt{62}$$

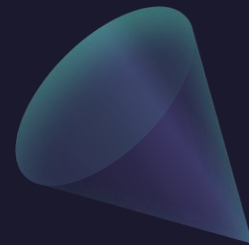
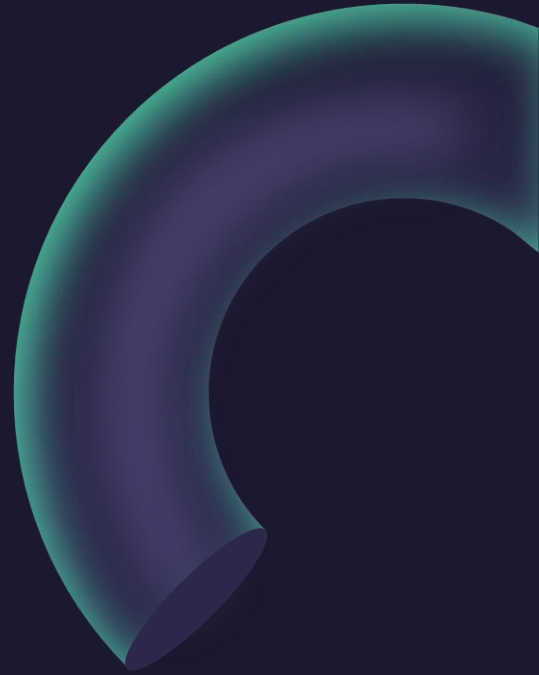
$$r'(2) = (1, 5, 6)$$

$$T(2) = \frac{r'(2)}{|r'(2)|} = \frac{1}{\sqrt{62}} (1, 5, 6)$$

$$= \left(\frac{1}{\sqrt{62}}, \frac{5}{\sqrt{62}}, \frac{6}{\sqrt{62}} \right)$$



Example: Find the derivative of the function given by
$$r(t) = (1 + t, 2 + 5t, -1 + 6t)$$
and find the unit tangent vector at $t = 2$.



Example: Find the derivative of the function given by

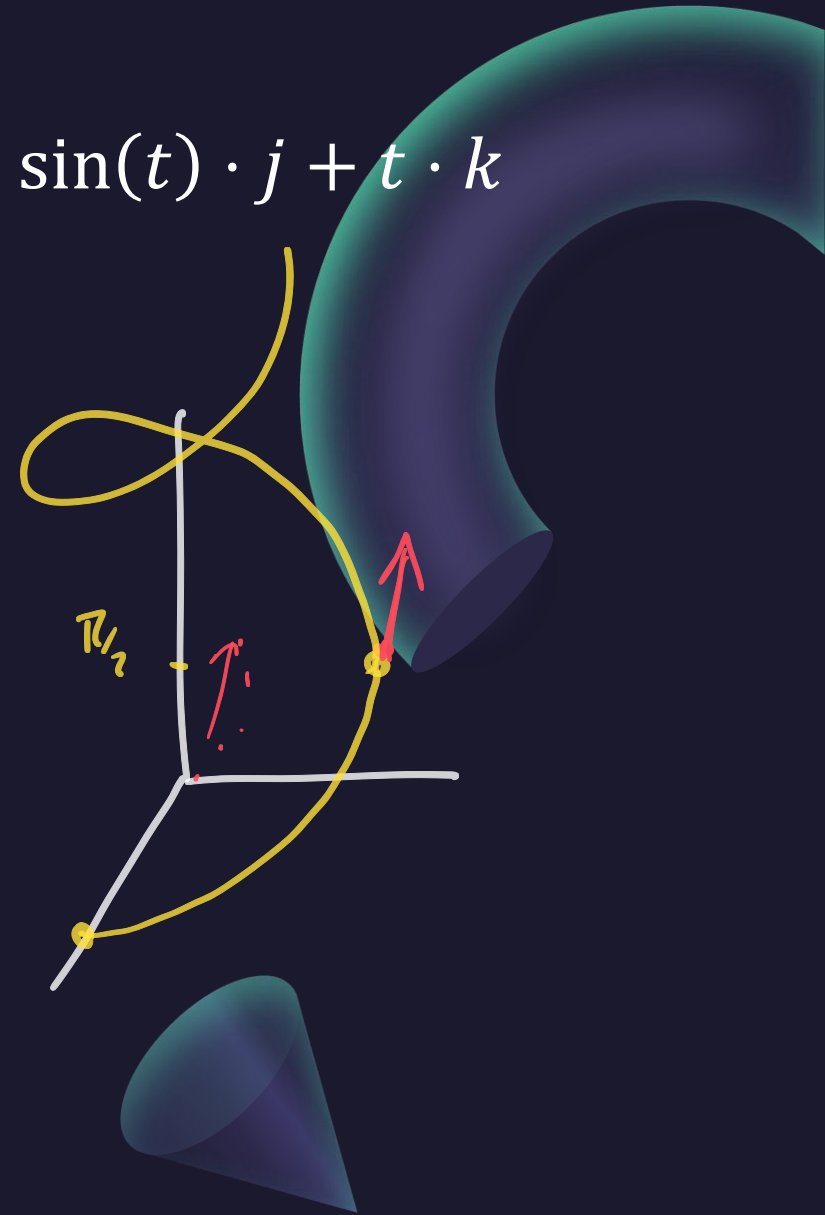
$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$
and find the unit tangent vector at $t = \frac{\pi}{2}$.

$$r'(t) = (-\sin(t), \cos(t), 1)$$

$$t = \pi/2 \quad r'(\pi/2) = (-1, 0, 1)$$

$$r(\pi/2) = (0, 1, \pi/2)$$

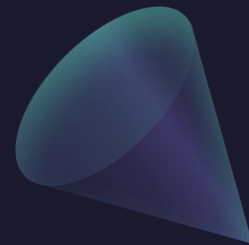
$$T(\pi/2) = \frac{r'(\pi/2)}{|r'(\pi/2)|} = \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$$



Example: Find the derivative of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

and find the unit tangent vector at $t = \frac{\pi}{2}$.



Differentiation Rules for Vector-valued Functions

Suppose $\mathbf{u}$ and $\mathbf{v}$ are differentiable vector functions, c is a scalar, and f is a real-valued function.

Then

$$1. \frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$$

$$2. \frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$$

$$3. \frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t) \leftarrow \text{Prod Rule}$$

$$4. \frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t) \quad \text{DOT PRODUCT!}$$

$$5. \frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$$

$$6. \frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t)) \quad (\text{Chain Rule})$$

If $|\mathbf{r}(t)| = c$ (a constant), then $\mathbf{r}'(t)$ is orthogonal to $\mathbf{r}(t)$ for all t .

Integrals of “Vector-valued Functions”

$$\begin{aligned}\int_a^b \mathbf{r}(t) dt &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbf{r}(t_i^*) \Delta t \\ &= \lim_{n \rightarrow \infty} \left[\left(\sum_{i=1}^n f(t_i^*) \Delta t \right) \mathbf{i} + \left(\sum_{i=1}^n g(t_i^*) \Delta t \right) \mathbf{j} + \left(\sum_{i=1}^n h(t_i^*) \Delta t \right) \mathbf{k} \right]\end{aligned}$$

and so

$$\int_a^b \mathbf{r}(t) dt = \left(\int_a^b f(t) dt \right) \mathbf{i} + \left(\int_a^b g(t) dt \right) \mathbf{j} + \left(\int_a^b h(t) dt \right) \mathbf{k}$$

Example: Find the integral

$$\int_0^{\pi/2} r(t) dt$$

of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

$$\int_0^{\pi/2} r(t) dt = \left(\int_0^{\pi/2} \cos t dt, \int_0^{\pi/2} \sin t dt, \int_0^{\pi/2} t dt \right)$$

$$= \left(\sin t \Big|_0^{\pi/2}, -\cos t \Big|_0^{\pi/2}, \frac{t^2}{2} \Big|_0^{\pi/2} \right)$$

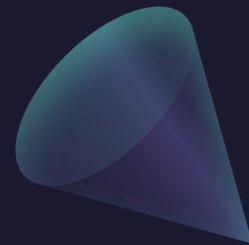
$$= \left(1, 1, \frac{(\pi/2)^2}{2} \right)$$

Example: Find the integral

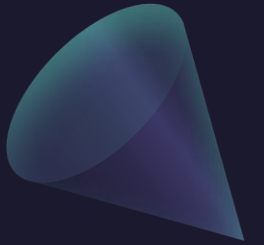
$$\int_0^{\pi/2} r(t) \, dt$$

of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

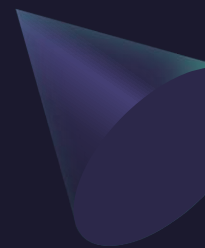


Questions?





ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

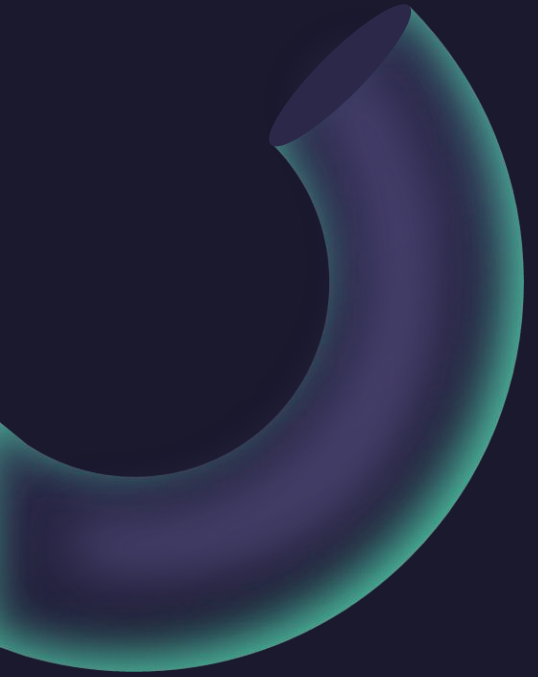
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Arc Length

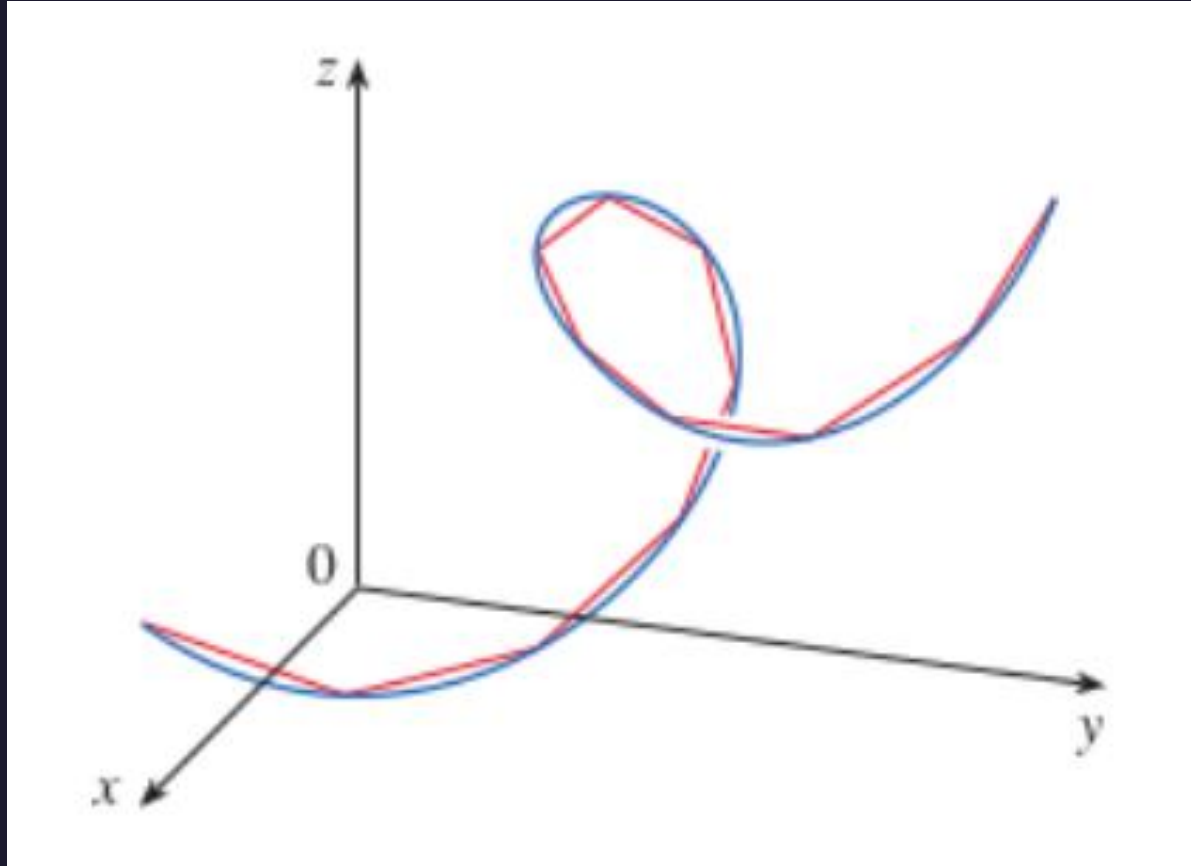


A decorative graphic consisting of a sphere and a cube, both rendered with a blue-to-teal gradient and a 3D effect, positioned in the upper-left corner of the slide.

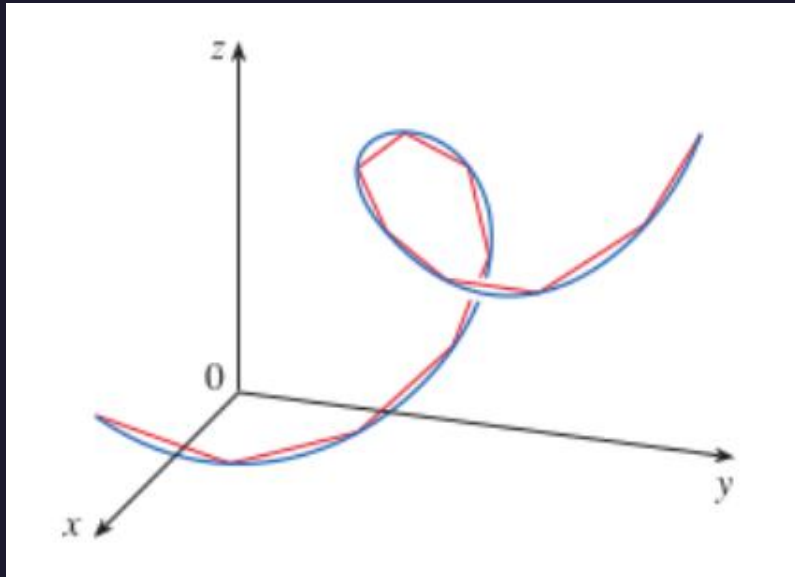
Today – Arc Length

- 
- A decorative graphic of a torus (donut shape) with a blue-to-teal gradient and a 3D effect, positioned on the left side of the slide.
- Length of a segment of a curve...
 - In the plane
 - In 3d-space

How do we compute the length of a curve?



How do we compute the length of a curve?



$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\begin{aligned} L &= \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt \\ &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \end{aligned}$$

How do we compute the length of a curve?

$$L = \int_a^b |\mathbf{r}'(t)| dt$$

because, for plane curves $\mathbf{r}(t) = f(t) \mathbf{i} + g(t) \mathbf{j}$,

$$|\mathbf{r}'(t)| = |f'(t) \mathbf{i} + g'(t) \mathbf{j}| = \sqrt{[f'(t)]^2 + [g'(t)]^2}$$

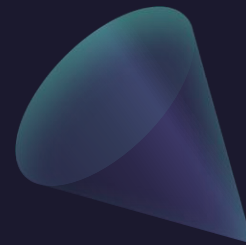
and for space curves $\mathbf{r}(t) = f(t) \mathbf{i} + g(t) \mathbf{j} + h(t) \mathbf{k}$,

$$|\mathbf{r}'(t)| = |f'(t) \mathbf{i} + g'(t) \mathbf{j} + h'(t) \mathbf{k}| = \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}$$

Example: Find the length of the curve given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

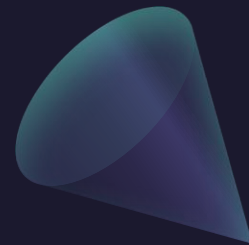
from the point $P = (1, 0, 0)$ to the point $Q = (1, 0, 2\pi)$.



Example: Find the length of the curve given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

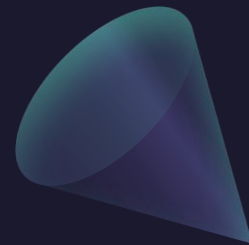
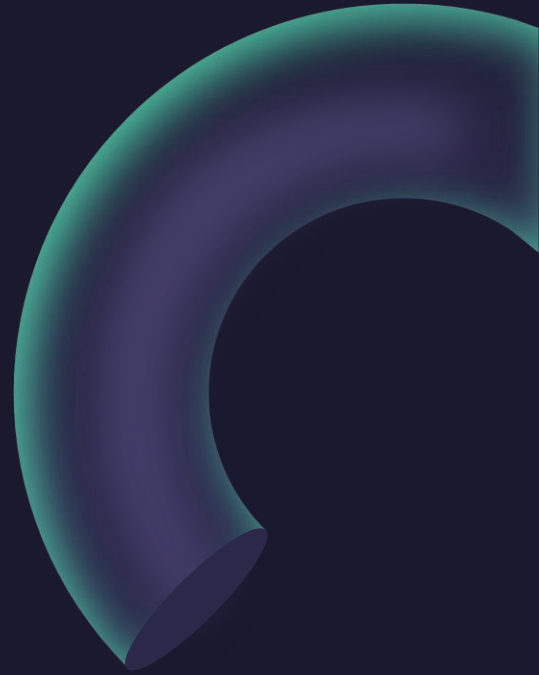
from the point $P = (1, 0, 0)$ to the point $Q = (1, 0, 2\pi)$.



Example: Find the length of the curve given by

$$r(t) = \left(2t, t^2, \frac{t^3}{3}\right)$$

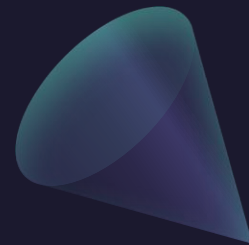
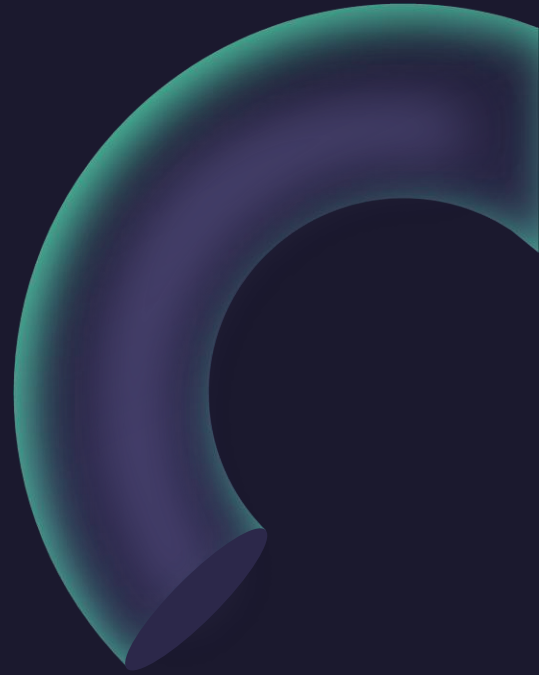
from $t = 0$ to $t = 1$.



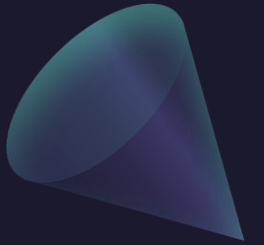
Example: Find the length of the curve given by

$$r(t) = \left(2t, t^2, \frac{t^3}{3}\right)$$

from $t = 0$ to $t = 1$.



Questions?



Thank you

Until next time.

